C++ CASE STUDY
Currency Exchange & Forex Simulation System

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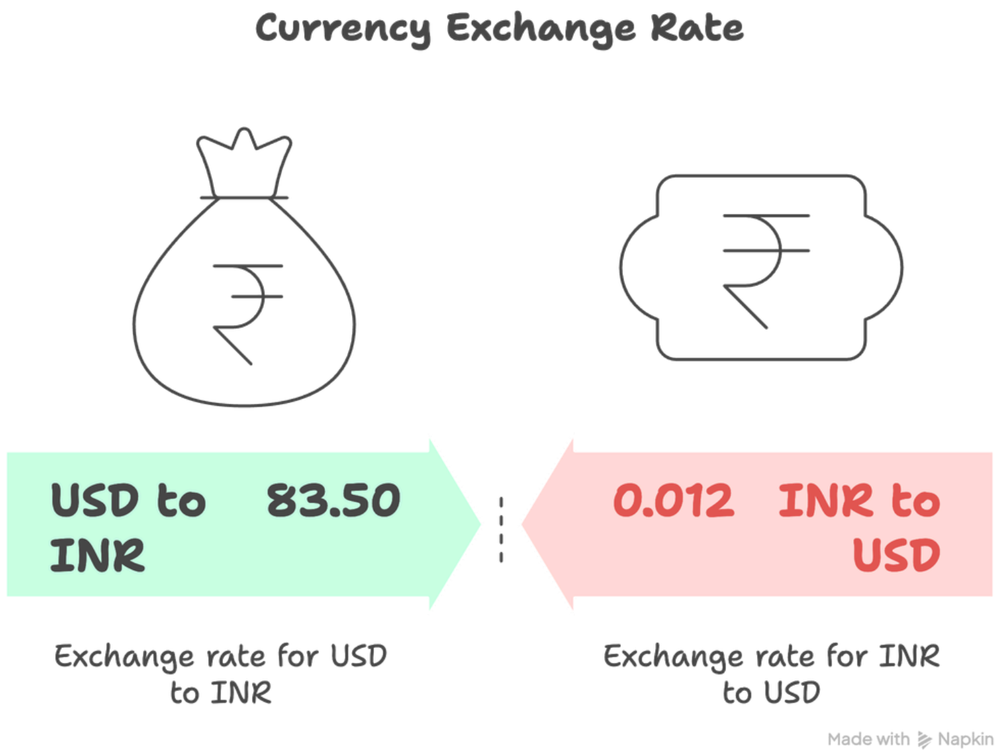
PROBLEM STATEMENT

Currency conversion requires accurate exchange rates and proper calculation. Manual conversion can lead to errors and confusion.

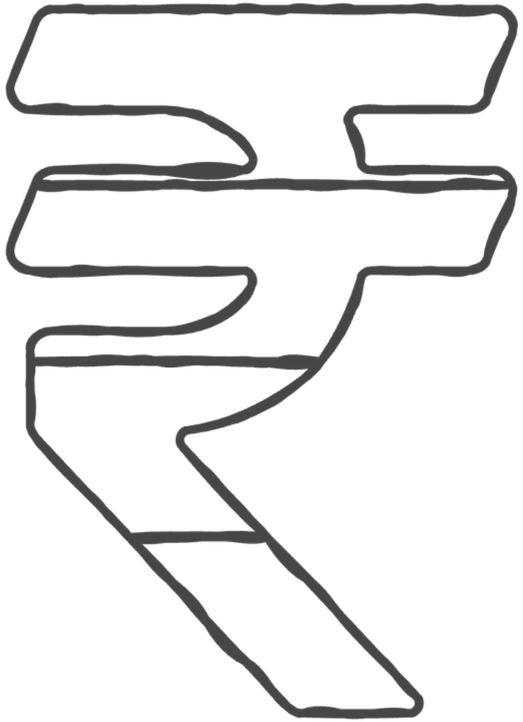
GOAL OF THE SYSTEM :

- Manage exchange rates
- Convert currencies
- Maintain transaction history



SOLUTION

Currency Exchange System Overview



Real-time Simulation

Mimics actual currency trading scenarios

Multi-Currency Support

Facilitates transactions across various currencies

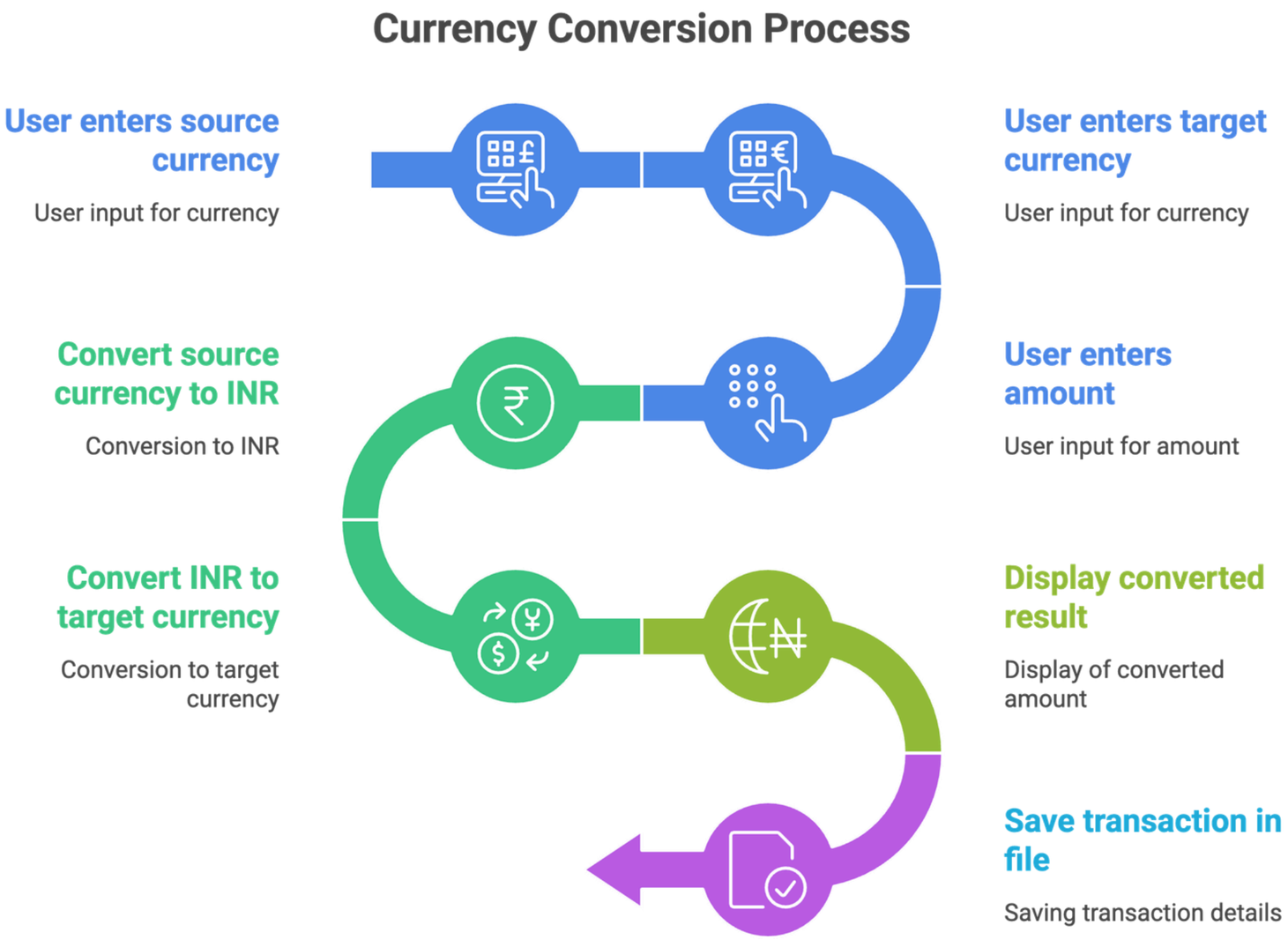
Rate Management

Keeps exchange rates up-to-date

Transaction Records

Stores detailed history of all exchanges

Algorithm / Working Logic




```
===== Forex Simulation System =====
1. Show Exchange Rates
2. Update Exchange Rate
3. Convert Currency
4. View Transaction History
5. Exit
Enter your choice:
```

Made with Napkin

CONCLUSION

1.Developed a basic Forex Currency Converter using C++.

2.Implemented features like rate viewing, updating, currency conversion, and transaction storage.

3.Applied core programming concepts like structures, functions, and file handling.

4.Demonstrates a simple simulation of foreign exchange systems.

```cpp
#include <iostream>
#include <fstream>
#include <ctime>
#include <algorithm>
using namespace std;

struct Currency {
    string code;
    double rateToINR;
};

Currency currencies[10] = {
    {"USD", 83.0},
    {"EUR", 90.0},
    {"GBP", 105.0},
    {"JPY", 0.55},
    {"AUD", 55.0},
    {"CAD", 60.0},
    {"CHF", 92.0},
    {"CNY", 11.5},
    {"SGD", 61.0},
    {"INR", 1.0}
};


string toUpperCase(string str) {
    transform(str.begin(), str.end(), str.begin(), ::toupper);
    return str;
}

int findCurrency(string code) {
    for(int i = 0; i < 10; i++) {
        if(currencies[i].code == code)
            return i;
    }
    return -1;
}

void showRates() {
    cout << "\n--- Current Exchange Rates (Base: INR) ---\n";
    for(int i = 0; i < 10; i++) {
        cout << "1 " << currencies[i].code
             << " = " << currencies[i].rateToINR
             << " INR\n";
    }
}

void updateRate() {
    string code;
    double newRate;

    cout << "Enter currency code to update: ";
    cin >> code;
    code = toUpperCase(code);

    if(code == "INR") {
        cout << "INR is base currency. Its rate cannot be changed!\n";
        return;
    }

    int index = findCurrency(code);

    if(index != -1) {
        cout << "Enter new rate (1 " << code << " = ? INR): ";
        cin >> newRate;

        if(newRate <= 0) {
```

THANK YOU

```
===== Forex Simulation System =====
1. Show Exchange Rates
2. Update Exchange Rate
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5. Exit
Enter your choice:
```

```
--- Current Exchange Rates (Base: INR) ---
1 USD = 83 INR
1 EUR = 90 INR
1 GBP = 105 INR
1 JPY = 0.55 INR
1 AUD = 55 INR
1 CAD = 60 INR
1 CHF = 92 INR
1 CNY = 11.5 INR
1 SGD = 61 INR
1 INR = 1 INR
```

```
Enter your choice: 2
Enter currency code to update: USD
Enter new rate (1 USD = ? INR): 91.61
Rate updated successfully!
```

```
Enter your choice: 3
Enter FROM currency code: usd
Enter TO currency code: inr
Enter amount: 4

4 USD = 366.44 INR
Transaction saved successfully!
```

forex.cpp

transactions.txt M

```
Date: Thu Mar  5 08:42:31 2026
From: USD
To: INR
Amount: 3
Converted: 249
------------------------
Date: Thu Mar  5 08:46:48 2026
From: AUD
To: USD
Amount: 2
Converted Amount: 1.5694
--------------------------------
Date: Thu Mar  5 09:47:22 2026
From: USD
To: INR
Amount: 4
Converted Amount: 366.44
--------------------------------
```